JINHENG ZHANG

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EDUCATION

University of Pennsylvania — Philadelphia, PA, USA

Ph.D. in Electrical and Systems Engineering — September 2024 – Present

Advisor: Prof. Shirin Saeedi Bidokhti

Tsinghua University — Beijing, China

B.Eng. in Electronic Information Science and Technology — September 2020 – June 2024

Advisor: Prof. Jintao Wang

RESEARCH INTERNSHIP

Yale University — New Haven, CT, USA

Research Intern, Yale Institute for Network Science — July 2023 – September 2023

Research Advisor: Prof. Leandros Tassiulas

PUBLICATIONS

[1] **Jinheng Zhang**, Tara Javidi, and Shirin Saeedi Bidokhti, "Sequential Rate–Distortion–Perception Trade-offs for Temporally Correlated Sources," in *Proceedings of the 2026 IEEE International Symposium on Information Theory (ISIT)*, pp. 1–6, 2026. [IEEE ISIT Student Travel Grant]

[2] **Jinheng Zhang**, Tara Javidi, and Shirin Saeedi Bidokhti, "Ultra-Low-Rate Semantic Text Compression with Anchored Language Models," accepted for presentation at *2026 IEEE Information Theory Workshop (ITW)*. [IEEE ITW Travel Grant] [Student best paper award candidate, to be presented]

[3] **Jinheng Zhang**, Tara Javidi, and Shirin Saeedi Bidokhti, "Rate–Complexity Trade-offs for Anchored Causal Exact Rejection Sampling," submitted to *2027 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*.

[4] Ali Zafari, **Jinheng Zhang**, Shirin Saeedi Bidokhti, and Shirin Jalali, "On the Optimality of Neural Compression for Structured Stationary Sources," accepted for presentation at *2026 Asilomar Conference on Signals, Systems, and Computers (ACSSC)*.

[5] Ruoyu Zhao, Yuting Chen, **Jinheng Zhang**, Zhehao Zou, and Tong Che, "Shared Gaussianization: What Gaussian Regularizers Certify About Contrastive Learning, and What They Miss," submitted to *2027 International Conference on Artificial Intelligence and Statistics (AISTATS)*.

[6] **Jinheng Zhang**, Haoyue Tang, Jintao Wang, Sastry Kompella, and Leandros Tassiulas, "Age Optimum Sampling in Non-Stationary Environment," *arXiv:2310.20275*, 2023.

SELECTED TECHNICAL SKILLS

Programming: Python (research experiments), MATLAB (simulation), C/C++

Research tools used: PyTorch, Transformers (Hugging Face), Sentence-Transformers, NumPy

SELECTED RESEARCH EXPERIENCES

Sequential Rate–Distortion–Perception Trade-offs for Temporally Correlated Sources

University of Pennsylvania, in collaboration with Prof. Tara Javidi (UC San Diego) *2025–2026*

- Formulated a sequential rate–distortion–perception framework for causal compression of stochastic sources, and characterized the minimum achievable sum rate over causal reconstruction kernels;
- Established structural reductions for Gauss-Markov sources under mean-squared error distortion and temporal Kullback–Leibler perception, and derived a parameterized optimization program for computing the optimal sequential trade-off;
- Compared sequential coding with the noncausal Gaussian-vector benchmark, quantifying how the causality penalty varies with temporal correlation, horizon length, and the stringency of the perception constraint, revealing how rate and distortion should be optimally allocated across time.

Ultra-Low-Rate Semantic Text Compression with Anchored Language Models

University of Pennsylvania, in collaboration with Prof. Tara Javidi (UC San Diego) *2026*

- Developed Anchored Channel Simulation with Language Models (ACSLM) for semantic text compression in ultra-low-rate regime, a two-stage framework in which informative anchor tokens condition a shared masked language model to generate candidate reconstructions;
- Evaluated rate–semantic-quality trade-offs on WikiText-103 and European Parliament corpora using cosine similarity and bidirectional entailment as semantic similarity metrics, demonstrating substantial gains in semantic quality in the ultra-low-rate regime over the considered baselines.

HONORS & FELLOWSHIPS

- 2026 IEEE ISIT Student Travel Grant
- 2026 IEEE Information Theory Workshop (ITW) Travel Grant
- 2024 Joseph A. Ganster, Jr. and Julia A. Ganster Engineering Fellowship, University of Pennsylvania
- 2023 Tsinghua Comprehensive Excellence Award (Tsinghua–Toyota Scholarship)
  - One of 32 recipients among 233 eligible students; awarded for outstanding performance in academics, research innovation, and public service
- 2022 Tsinghua Comprehensive Excellence Award (Tsinghua–Toyota Scholarship)
  - One of 34 recipients among 233 eligible students; awarded for outstanding performance in academics and research innovation
- Outstanding Member of 2022 Qihang Undergraduate Research and Innovation Program
  - One of 5 recipients among 233 eligible students; awarded for research excellence in Department of Electronic Engineering
- 2021 Tsinghua Academic Excellence Award
  - Academic ranking: 17th of 241 students in Department of Electronic Engineering